

Aryamaan (AJ) Jena

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Education

UCLA Anderson School of Management

PhD in Finance; Fink Center Award, Anderson Departmental Fellowship

Los Angeles, CA

2025 - 2030 (expected)

- Coursework: Asset Pricing Theory, Empirical Asset Pricing, Econometrics, Regression Analysis, Machine Learning Methods

University of Chicago

Chicago, IL

MS in Financial Mathematics (Financial Data Science); ASEAN Merit Scholarship

2021 - 2022

SP Jain School of Global Management

Mumbai and Sydney

BS in Data Science; Merit Scholarship; Dean's List (4 times)

2018 - 2021

Research

Conditioning or Orientation? Shrinkage Targets for Characteristic-Managed Portfolios

Working paper, 2026

- Showed with an eigenvector rotation test and permutation null that a characteristic-based shrinkage target gains out of sample mostly through conditioning (regularization), not characteristic information; 132 managed portfolios, 1963-2025.
- Benchmarked 23 estimators over 575 out-of-sample months: characteristic and scaled-identity targets tie for first by Sharpe ratio, significantly ahead of RIE, POET, RP-PCA and nonlinear shrinkage after Bonferroni correction.

Returns on Presale Housing Markets in India

Working paper, 2025

- Found repeat-sales presale returns of only 2-5% a year, earned mostly at launch, with no premium for holding through construction.

Research Experience

Indian Institute of Management Bangalore

Bengaluru, India

Research Associate, Prof. Venkatesh Panchapagesan

2024 - 2025

- Built repeat-sales return indices for presale housing by construction stage, builder reputation and housing segment across four cities.

Harvard Business School

Boston, MA

Research Associate, Profs. Victoria Ivashina and Josh Lerner

2023 - 2024

- Simulated mean-variance and Black-Litterman portfolios to estimate the cost of illiquidity in private equity and real estate.
- Built a panel of US financial firms; estimated the effect of acquisitions on FinTech innovation with synthetic control.

University of Chicago Booth School of Business

Chicago, IL

Research Assistant, Prof. Dacheng Xiu

2022 - 2023

- Predicted errors in Zillow's home price estimates with linear, regularized and nonparametric ML regressions.
- Built LendingClub default classifiers that became assignments and lecture material for Booth MBA students.

Industry Experience

Cboe Global Markets, Cboe Labs

Chicago, IL

Derivatives Strategy Analyst Intern; received a full-time return offer

Jun - Dec 2022

- Developed a quote filter for the VIX calculation (recomputed every 15 seconds) that uses put-call parity bounds to flag anomalous option quotes behind index spikes; shipped as an object-oriented Python module in the firm-wide quant library.
- Built a Python SEC EDGAR library that maps tickers, CUSIPs and SIC codes to one firm ID and parses 13F holdings; turned SEC/CFTC index eligibility rules into automated tests for iBoxx indices.
- Contributed unit tests and code reviews to the shared library (Jenkins, Kubernetes); migrated team workflows to AWS.

Pareto Frontier Capital Holdings

Chicago, IL

Quantitative Research Intern; feature engineering and PCA for crypto trading signals

Oct 2021 - Jan 2022

Teaching

UCLA Anderson, TA: Credit Markets (Prof. Pénasse); Venture Capital and Private Equity (Prof. Garmaise)

Fall 2026

Chicago Booth, TA: Financial Econometrics (Prof. Russell); Decoding FinTech (Prof. Xiu)

2022

Projects

Prediction Market Price Discovery

2026

- Collected 640,000+ Polymarket and Kalshi markets; matched 759 identical contracts with TF-IDF text similarity.
- Price gaps close with a half-life of about 2 hours; Polymarket leads price discovery (Gonzalo-Granger share about 0.7).

Skills

Statistics: Shrinkage estimation, factor models, PCA, regularization, gradient boosting, Bayesian inference, synthetic control

Finance: Empirical asset pricing, portfolio construction (mean-variance, Black-Litterman), credit risk, prediction markets

Tools: Python (pandas, scikit-learn, statsmodels, LightGBM), R, SQL, Git, AWS, SGE cluster, WRDS, EDGAR